

Privacy-preserving In-Context Learning for LLMs

Main Paper: Privacy-preserving In-Context Learning for LLMs,
[Wu et al. \[2024\]](#)

Summarizing paper & replicating subset of experiments with
open-source Gemma 2b.

Contents

0.1	Objective	2
0.2	Key ideas and methods	2
0.3	Foundations	3
0.3.1	Setup	3
0.4	Algorithms	4
0.4.1	Noisy-Max text classification	4
0.4.2	Embedding Space Aggregation	5
0.4.3	Keyword Space Aggregation	6
0.5	Results	7

0.1 Objective

This paper proposes a new approach for privacy-preserving in-context learning (ICL) with large language models (LLMs). The goal is to enable few-shot prompting on a sensitive dataset while guaranteeing that the prompt construction procedure satisfies differential privacy (DP), so that individual training examples stay private.

The method constructs example-query pairs and aggregate the outputs via a DP procedure built on an ensemble of LLMs. Each ensemble member is conditioned on a disjoint subset of private exemplars and proposes candidate few-shot demonstrations. These candidates are then combined using a DP selection/aggregation step (e.g., a noisy voting mechanism), producing a final prompt that can be used safely for ICL.

Conceptually, this “DP-ICL” paradigm focuses on: (i) **private few-shot example construction** via DP aggregation over many weak, disjoint views of the data, and (ii) **task inference** by an LLM using the resulting private few-shot examples. The authors evaluate the approach on both classification and text generation tasks and report that it can retain much of the accuracy of standard ICL while offering a formal privacy guarantee.

0.2 Key ideas and methods

Prompt leakage attacks can be used to recover the sensitive examples used in system prompting for in-context learning. Authors proposed a framework to integrate differential privacy into the process. This framework has roughly three stages.

1. Using Poisson sampling, partition the sensitive dataset before feeding it into the LLM model. Each partition consist of a set of examples that will be considered as user prompts for the LLM.
2. For each partition, pair the examplers with the queries. In this stage, we also use the query from the user. However, note that we don’t return an output at this stage.
3. Then these example - query pairs are fed to the LLM model.
4. Finally, all the outputs from all queries aggregated as a single answer. But in this stage they propose using DP mechanisms. Mainly they consider using Gaussian mechanism or Exponential mechanism.

The main component of this framework is actually the final stage. Differentially private sampling stage ensures the sensitive information is protected and not tracable. Even if the user is malicious and query is harmful, the final output will ensure the protection of sensitive examplers.

In detail, the authors suggest injecting noise in three possible ways.

- For text classification tasks, one can output the maximum vote count with Report-Noisy-Max algorithm [Dwork et al., 2014, Zhu and Wang, 2022]. However, this method is particularly designed for classification tasks with finite number of classes. For more general language tasks, one needs better methods.

Report-Noisy-Max algorithm has different versions, but the authors consider Gaussian noise for the ease of use and interpretability of Gaussian privacy.

- For more general tasks, one can focus on embedding space for the LLM outputs, which is called Embedding Space Aggregation. Simply adding Gaussian noise to the aggregate of embedded outputs will provide sufficient privacy guarantees. This method is unfortunately limited by how well the embedding representation preserves semantics and how reliably one can decode a (noisy) aggregated embedding back into a coherent output text.
- Finally, to decrease the dependency on embedding space representations, one can use the Keyword Space Aggregation. Similar to the Report-Noisy-Max, one chooses required number of words privately based on their histogram counts. However, this time the histogram is built using the output sentences of the queries. In this document, I skip the PTR for the sake of simplicity but the details can be found in the main document.

0.3 Foundations

0.3.1 Setup

Notation:

D : Private data

Q : Query

$o_i(Q)$: Output of query i

E : Embedding vectors

N : Number of subsets

Definition 1 (In-context learning (ICL) Wu et al. [2024]) *In-context learning (ICL) is the procedure of solving a task by providing a language model with a prompt that contains a small set of demonstration examples. Using the same notation, consider query-answer pairs $S := \{(Q_1, A_1), \dots, (Q_k, A_k)\}$. Then we generate next token via $\arg\max_A LLM(A|S + Q)$.*

Definition 2 (ϵ -Differential Privacy Dwork [2008], Dwork et al. [2014]) *A randomized algorithm M satisfies ϵ -differential privacy if for all datasets D_1, D_2 , which satisfy $\|D_1 - D_2\|_H = 1$, and $S \subseteq \text{Range}(M)$*

$$\Pr[M(D_1) \in S] \leq e^\epsilon \times \Pr[M(D_2) \in S]$$

where $\|D_1 - D_2\|_H = \sum_{i=1}^N 1_{D \neq D'}$

Definition 3 (Rényi Differential Privacy (RDP) Mironov [2017]) *Let P and Q be distributions with P absolutely continuous with respect to Q . For order $\alpha > 1$, the Rényi divergence is*

$$D_\alpha(P\|Q) = \frac{1}{\alpha - 1} \log \mathbb{E}_{x \sim Q} \left[\left(\frac{P(x)}{Q(x)} \right)^\alpha \right].$$

A randomized mechanism M satisfies (α, ϵ) -RDP if for all neighboring datasets $D \sim D'$, we have

$$D_\alpha(M(D) \| M(D')) \leq \epsilon.$$

Definition 4 (Privacy loss random variable [Dwork et al. \[2014\]](#)) Fix two neighboring datasets $D \sim D'$. For any output o in the range of M , define the privacy loss at o as

$$\ell_M^{D,D'}(o) := \log \frac{\Pr[M(D) = o]}{\Pr[M(D') = o]}.$$

The corresponding privacy loss random variable is $L_M^{D,D'} := \ell_M^{D,D'}(O)$ where $O \sim M(D)$.

A mechanism M is ϵ -differentially private if and only if for all neighboring $D \sim D'$, we have $\ell_M^{D,D'}(o) \leq \epsilon$ for all outputs o , or (ϵ, δ) -DP when $\Pr[L_M^{D,D'} > \epsilon] \leq \delta$. Note that for approximate DP, only one way is correct. The bound on privacy loss random variable implies privacy but the reverse direction is not correct.

0.4 Algorithms

Here, I present the main algorithms from the paper and related proofs.

0.4.1 Noisy-Max text classification

Algorithm 1 DP-ICL via RNM-Gaussian

Input: Private data D , query Q , model LLM, noise σ , N .

Output: Privately selected top-1 bin from H .

Partition $D_1, D_2, \dots, D_N$

for $i \in \{1, \dots, N\}$ **do**

 Form pairs $D_i^Q = D_i \cup \{Q\}$

 Obtain the model output $o_i(Q) = LLM(D_i^Q)$

 Convert $o_i(Q)$ to one-hot vector with equal lengths for each class $1, \dots, k$

Construct H from one-hot vectors

Add noise $\mathcal{N}(0, \sigma^2)$ to each entry of H to make it $\hat{H}$.

Return the argument with maximum count in $\hat{H}$.

Privacy guarantee

This simply boils down to Gaussian noise and privacy cost of Gaussian mechanism. Note that we will discuss the regular (ϵ, δ) -DP here, not the Gaussian Differential Privacy [[Dong et al., 2022](#)].

Theorem 1 Algorithm 1 is (ϵ, δ) -DP.

For the proof, we can use privacy loss random variable. Then first of all, we have the following lemma. For a proof of Lemma 1, see [Kamath](#).

Lemma 1 *Let $X, X' \in \mathcal{X}$ be neighboring datasets. Consider a randomized algorithm in a form $M(Y) = f(Y) + \mathcal{N}(0, \sigma^2 I)$. Then the privacy loss random variable (PRV) between $M(X)$ and $M(X')$ has normal distribution as follows.*

$$PRV \sim \mathcal{N}\left(\frac{\|f(X) - f(X')\|_2^2}{2\sigma^2}, \frac{\|f(X) - f(X')\|_2^2}{\sigma^2}\right)$$

Proof 1 (Proof of Theorem 1) *Algorithm 1 is an instance of Gaussian mechanism with Gaussian noise. So, we need to prove the DP property for Gaussian noise. For $Z \sim \mathcal{N}(0, 1)$*

$$PRV = \frac{\|f(X) - f(X')\|_2}{\sigma} Z + \frac{\|f(X) - f(X')\|_2^2}{2\sigma^2}$$

Then

$$\mathbb{P}[|PRV| \geq \epsilon] = \mathbb{P}\left[|Z| \geq \frac{\sigma\epsilon}{\|f(X) - f(X')\|_2} - \frac{\|f(X) - f(X')\|_2}{2\sigma}\right]$$

Now, plug $\sigma = \frac{\Delta_2 t}{\epsilon}$, and use Gaussian tail bound with $t = \sqrt{2 \log(2/\delta)}$ to reach the DP result.

0.4.2 Embedding Space Aggregation

We now present Embedding Space Aggregation (ESA) for language generation [Wu et al., 2024].

Algorithm 2 DP-ICL via Embedding Space Aggregation (ESA)

Input: Private data D , query Q , model LLM, noise σ , number of subsets N , public candidate sentences $\mathcal{C} = \{c_1, \dots, c_M\}$.

Output: Privately aggregated sentence $\hat{s}$.

Partition $D_1, D_2, \dots, D_N$.

for $i \in \{1, \dots, N\}$ **do**

Form exemplar-query pair $D_i^Q = D_i \cup \{Q\}$.
 Obtain model output sentence $s_i = \text{LLM}(D_i^Q)$.
 Project s_i into an embedding vector $e_i \in \mathbb{R}^d$.

Compute the mean embedding $\bar{e} = \frac{1}{N} \sum_{i=1}^N e_i$.

Add Gaussian noise to obtain $\tilde{e} = \bar{e} + \mathcal{N}(0, \sigma^2 I_d)$.

Return $\hat{s} = \arg \max_{c \in \mathcal{C}} \cos(\text{Emb}(c), \tilde{e})$.

Privacy Guarantee

Theorem 2 *Algorithm 2 is (ϵ, δ) -DP.*

Proof 2 (Proof of Theorem 2) *Proof directly follows from the proof of theorem 1 as this is an instance of Gaussian mechanism.*

0.4.3 Keyword Space Aggregation

Keyword Space Aggregation privately selects a small set of representative keywords by building a histogram over keywords extracted from the ensemble outputs and then applying a DP selection rule (e.g., noisy max / exponential mechanism) to pick the top keywords. Here is a pseudocode for Keyword Space Aggregation algorithm:

Algorithm 3 Keyword Space Aggregation (KSA) [Wu et al., 2024]

Input: Private data D , query Q , model LLM, noise σ , number of subsets N , public candidates $\mathcal{O}_C$ (size C), max token length M , method $\in \{\text{JEM}, \text{PTR}\}$.

Output: A private set of keywords/tokens (used to reconstruct the final answer).

Partition $D_1, D_2, \dots, D_N$.

for $i \in \{1, \dots, N\}$ **do**

Form exemplar-query pair $D_i^Q = D_i \cup \{Q\}$.
 Obtain model output sentence $o_i(Q) = \text{LLM}(D_i^Q)$.

For each token, count in how many sentences in $\{o_i(Q)\}$ it appears, and form a histogram H .

if $method = \text{JEM}$ **then**

Return $\text{JointEM}(M, H)$.

else

$\hat{k} = \text{FindBestK}(H)$.
 Return $\text{TopKwithPTR}(\hat{k}, H)$.

Joint Exponential Mechanism

Exponential mechanism is designed for releasing an object that would maximize a certain objective function. In this problem, keyword space aggregation method releases a set of top keywords that maximizes the LLM’s predictions. Instead of using exponential mechanism in a composition, joint exponential mechanism from Gillenwater et al. [2022] offers a better solution for choosing all k objects jointly, and no need for composition. The version in the paper uses an efficient sampling algorithm that leads to the complexity of $O(dk \log(k) + d \log(d))$ [Gillenwater et al., 2022].

Utility function for joint exponential mechanism is defined as follows. Let $(c_1, \dots, c_d)$ be the item counts sorted in nonincreasing order, and let $S = (s_1, \dots, s_k)$ be a candidate top- k sequence. Then

$$u^*(D, S) = \begin{cases} -\max_{i \in [k]} (c_i - c_{s_i}) & \text{if } s_1, \dots, s_k \text{ are distinct,} \\ -\infty & \text{otherwise.} \end{cases}$$

A nice property of u^* is as follows. This property implies that the objective utility function stays effective with high probability. This is a direct application of basic utility guarantee of exponential mechanisms (see [Dwork and Roth, 2014]).

Theorem 3 (Utility guarantee for JOINT [Gillenwater et al., 2022]) *Let $c_1, \dots, c_k$ denote the true top- k counts for dataset D , and let $\tilde{c}_1, \dots, \tilde{c}_k$ denote those output by JOINT. With probability at least $99/100$,*

$$\max_{i \in [k]} (c_i - \tilde{c}_i) \leq \frac{2[k \ln(d) + 5]}{\epsilon}.$$

Finally, we have the following theorem. This is true because algorithm 3 uses DP mechanism only at Joint EM or PTR part. Since Joint EM is an instance of EM with the given utility function, we have the DP property.

Theorem 4 *Algorithm 3 with Joint EM is ϵ -DP.*

0.5 Results

Overall, DP-ICL consistently improves over zero-shot prompting and, across tasks, stays close to the corresponding non-private baselines.

Text classification (RNM-Gaussian). The paper evaluates DP-ICL on multiple classification datasets (e.g., SST-2, Amazon, AGNews, and TREC) by privately aggregating an ensemble of few-shot predictions via Report-Noisy-Max with Gaussian noise. The reported accuracy drop from non-private aggregation to private aggregation is small at moderate privacy budgets (e.g., on SST-2 with GPT-3 Babbage, the performance at $\epsilon = 3$ is essentially unchanged relative to the non-private baseline), while even a conservative $\epsilon = 1$ often substantially outperforms zero-shot (e.g., improvements on the order of $\approx 20\%$ on AGNews are reported). The study also shows that stronger base models improve DP-ICL directly: for SST-2 at $\epsilon = 1$, GPT-3 Davinci is about 3.1% higher than GPT-3 Babbage.

Document question answering (ESA vs. KSA-by-PTR). For document QA, the paper compares embedding-space aggregation (ESA) and keyword-space aggregation (KSA) using PTR, reporting ROUGE-1, BLEU, and Levenshtein similarity. Private aggregation remains competitive with non-private aggregation even at $\epsilon = 1$, and the privacy-induced degradation is minimal (e.g., the ROUGE-1 difference from $\epsilon = \infty(\text{Agg})$ to $\epsilon = 1$ is reported as $< 3\%$ for KSA-by-PTR and about 0.19% for ESA). KSA-by-PTR substantially outperforms ESA and can even exceed standard 1-shot ICL, which the authors attribute to the robustness of aggregating consensus keywords.

Dialog summarization (SAMSum). On SAMSum, the paper evaluates ESA and KSA with both PTR and Joint Exponential Mechanism (JointEM), using 4-shot prompting and ROUGE metrics. Under $\epsilon = 1$, the strongest method (KSA + JointEM) improves over zero-shot by roughly 4.5% (ROUGE-1), 3.0% (ROUGE-2), and 3.4% (ROUGE-L), while remaining within about $\sim 1\%$ (on average) of non-private aggregation.

Ablations and scalability. The paper reports that DP-ICL performance is relatively stable as the number of private queries grows (with less than 2% degradation observed up to 10,000 queries at $\epsilon = 3$ in their study), and it analyzes trade-offs with ensemble size and subsampling rates. They also provide cost/accuracy analyses suggesting that reasonable subsampling rates can keep API cost low while maintaining competitive accuracy.

Future direction (open-source evaluation). A natural next step is to replicate a subset of these experiments with an open-source model such as Gemma 2B, to test whether more capable open models change the privacy-utility trade-offs and to make the DP-ICL methods easier for the community to reproduce and extend.

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